Calculus 1 Stack 2 8.11.2026

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1. $A=\left\{\frac{p}{q}\in\mathbb{Q} \mid p^2<5q^2,\ p,q>0\right\}$

$\frac{p^2}{q^2}<5 \rightarrow \frac{p}{q}<\sqrt{5} = \sup(A)$

$\frac{p}{q}>0 = \min(A)$

2. $A=\left\{\frac{1}{5}+\frac{n}{5n+1} \mid n\in\mathbb{N}\right\}$

$\frac{n}{5n+1}<\frac{n}{5n}=\frac{1}{5}$

$\forall a\in A \quad 0<a<\frac{2}{5}$

3. $A=\left\{\frac{1}{2n^2}+\frac{1}{m} \mid m,n\in\mathbb{Z},\ m,n\neq 0\right\}$

$-1\le\frac{1}{m}\le 1 \quad \max(A)=\frac{3}{2}$

$0<\frac{1}{2n^2}\le\frac{1}{2} \quad \inf(A)=-1$

4. $A=\{x-y \mid -70<x<-42,\ 75<y<84\}$

$\min(A)=\min(x)-\max(y)=-154$

$\max(A)=\max(x)-\min(y)=-117$

5. $A=\left\{\frac{6m}{3n+6m} \mid m,n\in\mathbb{N}\right\}$

$\min(A)=\max(n),\ \min(m)=0$

$\max(A)=\max(m),\ \min(n)=\inf(A)=1$

6. $A=\{x\in\mathbb{R} \mid x<0,\ 18x^2+62x-40<0\}$

$9x^2+31x-20<0$

$(9x-5)(x+4)<$

$x_1=\frac{5}{9},\ x_2=-4 \quad -4<x<\frac{5}{9}$

7. $A=\left\{\sqrt{11n}-\sqrt{11}\cdot\lfloor\sqrt{n}\rfloor \mid n\in\mathbb{N}\right\}$

$a=\sqrt{11}\left(\sqrt{n}-\lfloor\sqrt{n}\rfloor\right)$

$0\le a<\sqrt{11}$